

India's quantum leap

HOW TCS AND LTIMINDTREE ARE BUILDING A HOMEGROWN QUANTUM FUTURE

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In the southern state of Andhra Pradesh, a quiet revolution is underway. Amaravati Quantum Valley, the nascent epicentre of India's quantum ambitions, is being shaped not just by government policy, but by two of India's heavyweight IT heavyweights: Tata Consultancy Services (TCS) and LTIMindtree.

These companies are calibrating qubits, crafting algorithms, and doing something few thought plausible a decade ago: building cutting-edge quantum computing platforms in India, for the world.

Dawn of the qubit era

For TCS, the commitment is both literal and local. "As a founding partner of the Amaravati Quantum Valley Tech Park, Tata Consultancy Services (TCS) is proud to be at the forefront of advancing quantum innovation in India," says Anil Sharma, Global Head of the TCS Co-Innovation Network (COIN).

That front row seat also comes with a front-row machine: a 156-qubit IBM Heron processor that Sharma calls "a transformative step for quantum research in the region." He adds that

this system offers "capabilities in qubit stability, coherence, and scalability – enabling researchers and industry partners to tackle real-world challenges that were previously beyond reach, including breakthroughs in drug discovery, advanced materials, and complex risk optimization".

The keyword here is access – something Sharma believes is as vital as infrastructure. "At TCS, we aim to

democratize quantum access for start-ups, researchers and students across India through academic collaborations and cloud access programs," he says.

This democratic ideal has a very real implication: India isn't just building a quantum machine; it's building a quantum community.

On the other end of the quantum corridor is LTIMindtree, a firm with serious translational ambi-

tion. "LTIMindtree envisions playing a catalytic role in India's quantum future," says Indranil Mitra, Vice President of LTIMindtree Research. "Our long-term vision for quantum computing centers



INDRANIL MITRA
Vice President,
LTIMindtree Research

on accelerating industrial adoption by bridging the gap between scientific breakthroughs and real-world problems". The company's

approach is refreshingly pragmatic. "We've launched the Quantum Jumpstart Program, a structured

pathway for organizations to begin their quantum journey – right from awareness to proof-of-concept and implementation," says Mitra.

While TCS is laying out the cloud-powered infrastructure, LTIMindtree is handing out the boarding passes to India Inc., ushering businesses into a future powered by quantum-enhanced decision making.

It isn't just R&D

If all this sounds like corporate research theatre – well-dressed ambition with little follow-through – rest assured, these companies are clocking real progress.

Take LTIMindtree's partnership with IBM Quantum Network. "As the first Indian global system integrator to join the IBM Quantum Network, LTIMindtree gained premium access to IBM's cutting-edge quantum systems – including Heron processor-powered systems – and a sophisticated software development stack," says Mitra. This



ANIL SHARMA
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